



Department of Computer Science and Engineering
Midterm Examination Spring 2023
CSE 321: Operating Systems

Duration: 1 Hour 15 Minutes

Total Marks: 25

Answer the following questions.
Figures in the right margin indicate marks.

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1. a) **Explain** why main memories are usually volatile. [2]
CO1 b) **Distinguish** between monolithic and microkernel OS structures. [3]
c) A program has a process that will allow its child process to complete first. **Which** system call can be used in this scenario? **Explain** what may happen in absence of this system call. [2]
d) **Find** the output of the following code snippet. [3]

```
const int len = 2;
int main(){
    int id;
    int a[] = {10,7};
    int b = len-1;
    id = fork();
    if (id < 0){
        printf("fork failed\n");
    }
    else if(id == 0){
        printf("child process executing\n");
    }
    else{
        wait(NULL);
        printf("parent process executing\n");
        a[b-1]=a[b-1]-2;
        a[b] = a[b]+2;
    }
    for(int i=0;i<len;i++){
        printf("value      of      a[%d]:
%d\n",i,a[i]);
    }
    return 0;
}
```

2.
CO2

Processes	Arrival Time	Burst Time
P1	0	11
P2	20	6
P3	14	9
P4	20	8
P5	15	8
P6	16	8
P7	2	2

a) **Draw** a Gantt chart and illustrate the execution of the process using the **Round Robin** scheduling algorithm (**time quantum = 5 units**). **Calculate** the **average waiting** and **turnaround time**,

[3+2]

b) **Apply Shortest Remaining Time First (SRTF)** scheduling algorithm. **Draw** the Gantt chart and **Calculate** the **average waiting** and **turnaround time**.

[2+2]

c) **Compare** the results and **identify** the most suitable scheduling algorithm in this scenario.

[1]

3. a) A system has processes to execute of which **35%** is parallel. If the number of cores is increased from **3 to 5**, Explain what will be the increase/decrease in performance? [2]
- CO3**
- b) A program has multiple threads that need to perform a task which involves similar computation on a large scale of data. **Identify** which kind of parallelism can be applied in this scenario. **Provide** proper justification with a **real-life example** of the scenario mentioned above. [3]